ÇAĞLA AKIN

London, UK • github.com/cagla-akin

Causal Inference & Experimentation · Machine Learning · Amazon · Carnegie Mellon University (PhD) · Imperial College London (Data Science)

SUMMARY

I am a problem solver first. Over 7 years at Amazon, my job was to take on whatever stood between the business and its goals: I built tools that removed inefficiencies, unlocked savings, and supported multi-billion-dollar planning across the global supply chain, partnering with science, engineering, product, and finance teams. My solutions stand on analytical grounds. With PhD-level training in causal inference from Carnegie Mellon University and Executive Education in Data Science and ML from Imperial College London, I break ambiguous questions down into a structured framework, build the solution, and productionize it with SQL-based ETL pipelines that keep it running on updated data without manual effort. I take ownership end to end, hold myself to a high bar for delivery, and adapt quickly to new domains and shifting priorities. I hold Tech Nation UK's Exceptional Talent endorsement in Digital Technology and do not require sponsorship.

EXPERIENCE

AMAZON.COM, Inc, | Seattle, US & London, UK — Aug 2019 - Present

Finance Manager, Supply Chain Optimization Technologies

- Built a supply-chain simulation model from scratch in two weeks during COVID-19, replacing fragmented cross-system forecasts with a unified model of backlog clearance against labor capacity across NA, EU, and Japan; engineered ETL pipelines and daily regional metrics that became the single source of truth behind the SVP decision to cut promised delivery times by 76% (21→5 days), contributing to 38% revenue growth in 2020.
- Developed Amazon's North America 1-Day delivery-speed forecasting model; engineered predictive features across inventory placement and transportation and validated via backtesting with 197 bps average error vs a ±500 bps threshold, establishing the model as leadership's decision-making foundation for pursuing a $3.8B revenue opportunity.
- Architected the end-to-end data infrastructure for Amazon's transportation cost anomaly detection product that uncovered $300M in free cash flow; co-designed its causal-inference model with Core AI economists, engineered model-ready features, and used SQL and Python to integrate 50+ tables across 7 source clusters into a curated 300GB data library powering the tool's analytics.
- Partnered with an economist on a fixed-effects causal study of labor forecast error, then built the cost-and-savings model and the SQL data pipelines behind it; automated ETL reporting into weekly business reviews, surfacing $78M in annual savings and realizing $11.6M as forecast error fell 4.2%.
- Built the first metrics framework for profit lost to capacity shortfalls, and defined the outcome metric for an A/B test with a 10% inventory holdout that quantified $40M in lost Q4 sales, driving changes to the inventory-health engines.
- As acting Senior Product Manager, architected a global facility dataset consolidating fragmented upstream sources, cutting manual workloads 60% and saving $0.5M annually.
- Owned financial strategy for $10B inventory across EU and Japan; drove the finance-side forecast against the science team's Bayesian VAR model, auditing cost inputs to identify $50M in clearance opportunity and action $10M of it.

CARNEGIE MELLON UNIVERSITY | Pittsburgh, PA, US — Aug 2016 – May 2019

PhD Candidate/Doctoral Researcher

- Trained in causal inference and econometrics; applied difference-in-differences and regression-based quasi-natural experiment designs across three empirical papers on regulatory interventions in banking, municipal bond, and energy markets.
- Earned the American Accounting Association's Best Doctoral Student Paper Award; built novel research datasets via web scraping.

DELOITTE TOUCHE TOHMATSU | Istanbul, Turkey — Jun 2015 – Jul 2016

Financial Advisory Associate

- Advised C-suite executives on M&A decisions, translating financial due diligence and valuation analysis into recommendations that shaped buy- and sell-side negotiations.

EDUCATION

IMPERIAL COLLEGE BUSINESS SCHOOL | London, UK — Oct 2025 – Apr 2026

Executive Education in Data Science

CARNEGIE MELLON UNIVERSITY | Pittsburgh, PA, US — Aug 2016 – May 2019

Tepper School of Business, MSc. in Industrial Administration (Accounting)
(Completed PhD coursework and research before earning a Master's degree)

BILKENT UNIVERSITY | Ankara, TR — Sep 2011 – Jun 2015

Bachelor of Science in Business Administration, Minor in Economics

UNIVERSITY OF BOCCONI | Milan, IT — Jan 2014 – Jul 2014

Exchange Student, Finance and Economics

SKILLS

Causal Inference & Experimentation: Difference-in-differences, regression discontinuity design, A/B testing
Machine Learning & Statistics: Regression, classification, clustering, simulation, statistical modeling, supervised & unsupervised learning
Soft Skills: Executive communication, senior-stakeholder management, cross-functional partnership, project & program management
Tools: Python (pandas, NumPy, scikit-learn, statsmodels), SQL, AWS SageMaker, Git, Tableau, QuickSight, MS Excel

DATA SCIENCE PROJECTS & TRAINING

Applied AI Bootcamp, AllWomen Tech | London, UK

- Hands-on training in machine learning, deep learning, generative AI, and NLP using Python.
- Worked with pretrained LLMs (Llama 2, Hugging Face) for NLP tasks including text classification and summarisation; applied prompt engineering to adapt models for business use cases.
- Studied AI governance and responsible AI: Bias and fairness metrics, protected attributes, model explainability (SHAP, global vs local), and the regulatory landscape including the EU AI Act, GDPR compliance, model cards and dataset documentation.
- **Spotify Hit Predictor**: XGBoost model predicting song chart success from ~170K tracks; iterated linear regression → random forest → XGBoost (~80% accuracy), finding artist reputation matters far more than audio characteristics. Deployed to an AWS SageMaker inference endpoint and published as an interactive project site (link).
- **Suno AI Brand Monitor**: NLP brand-monitoring prototype analysing public conversations with VADER sentiment and keyword-guided topic modelling; found ~68% positive sentiment, with copyright and legal exposure as the key reputational risk.

AWARDS & RECOGNITION

- Amazon Future Leader (AFFL) Constellation Award | Amazon — Jul 2025
- Exceptional Talent in Digital Technology | endorsed by Tech Nation UK — Apr 2025
- Best Doctoral Student Paper Award | American Accounting Association — May 2019
- William Larimer Fellowship | Carnegie Mellon University (5-year doctoral fellowship) — Aug 2016

PUBLICATIONS

- Akin, C. (2025). Causal Inference in E-Commerce and Technology Business. Harvard Business Review Türkiye (link).
- Akin, C. (2019). Consolidation of Securitization Entities and the Effects on Charitable Lending. Journal of Management Policy and Practice, 20(4) (link).
- Akin, C. Financial Disclosure in Municipal Bond Markets. Working Paper.
- Research contribution (acknowledged): Intended and Unintended Consequences of Facility-Level Mandatory CSR Disclosure. Journal of Accounting, Auditing & Finance (2026) (link).

SPEAKING & TEACHING

- Guest Lecturer, Causal Inference Methods | Aalto School of Business / Helsinki GSE — Nov 2024
- Panelist & Moderator, NextGen Finance Transformation Conference | London — Jun 2026
- Judge, McGinnis Venture Competition | Carnegie Mellon University — Jan 2025, 2026
- Guest Speaker, career events | Bilkent University and Koç University, Türkiye — Mar 2025
- Moderator & Presenter, Capital Markets Research Session | Mid-Atlantic Regional Meeting — May 2019
- Presenter, Annual Accounting Mini Conference | Carnegie Mellon University — 2016, 2017, 2018

LEADERSHIP & COMMUNITY

- **Product Owner, Code Your Future (CYF Dashboard)**: Lead a technical team of 9 (Agile/Scrum) shipping an education platform serving 300+ underrepresented trainees and volunteers.
- Tech Startup Mentor, ITU Çekirdek: Advise seed-stage tech start-ups on digital strategy and operational scaling in weekly mentoring sessions.
- Young Enterprise UK: Coach high-school enterprise teams through the programme year and judge the national final, assessing the top companies among 50+ competing teams.
- DJ & Producer, Sunrise Fiancée: Curate and perform melodic house and techno across London radio, club, and hospitality settings, including a residency at Hart Shoreditch Hotel; programme live sets to venue, audience, and moment.